

KAAN BOREKCI

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EDUCATION

Northwestern University

Evanston, IL

Master of Science in Mechanical Engineering | GPA: 4.0/4.0

Expected Jun 2026

Bachelor of Science in Biomedical Engineering | Major GPA: 3.7/4.0 | GPA: 3.5/4.0 | Dean's List

Jun 2025

WORK EXPERIENCE

Shirley Ryan Ability Lab, Center for Bionic Medicine | Arun Jayaraman Lab

Chicago, IL

Biomedical Machine Learning & Feature Engineering Intern

May 2024 – Present

- Pioneered a pipeline to assess 8+ MMPose 3D pose lifting models by processing 17 uncalibrated infant videos
- Increased feature generation by 50% through depth-based 2D-to-3D reconstruction of pseudo-ground truth sets
- Engineered an evaluation framework applying 10+ metrics across 17 keypoints per frame, validating predictions with clinically supported infant anatomical constraints (0.5, 1, and 6 months of age)
- Applied custom algorithm on 375k+ pseudo-ground truth frames with CUDA GPUs, identifying highest performing pose lifter for ML feature extraction, to enable diagnosis of infant motor delays up to 12 months earlier

International Institute for Nanotechnology | Chad Mirkin Lab

Evanston, IL

Nanotechnology Intern

Jun 2023 – Jul 2023

- Optimized crystallization conditions of DNA-coated colloidal nanoparticles, reducing formation time by 48+ hrs
- Introduced novel ImageJ workflow analyzing 80+ hrs of crystal data, demonstrating results to Dr. Mirkin and lab

Innovation and New Ventures Office at Northwestern | Dr. Lisa Dhar

Evanston, IL

Technology Transfer Intern

Jul 2022 – Aug 2022

- Built internal database tracking \$2.75B raised across 180+ startups, supporting tech-transfer strategy development
- Consolidated 12 yrs of scattered funding data into a guide used by the Associate VP of INVO to drive IP strategies

Querrey Simpson Institute for Bioelectronics | John Rogers Lab

Evanston, IL

Bioelectronics Research Intern

May 2020 – Sep 2020; Jul 2019 – Aug 2019

- Evaluated embedded sensors for post-graft skin flap monitoring and hydrocephalus shunt flow detection
- Spearheaded experiments on phantom models, simulating physiological flow conditions to test device accuracy
- Coauthored peer-reviewed *Nature* article with experimental contributions and figure designs in Adobe Illustrator

PROJECT EXPERIENCE

Diaphragm Control for Pediatric Congenital Central Hypoventilation Syndrome

Chicago, IL

Senior Capstone & Independent Project | Lurie Children's Hospital CAMP Division R&D Engineer

Sep 2024 – Present

- Selected to present in BMES 2025 (San Diego), patent pending with IRB-approved trials and IEEE paper underway
- Led 4-person team pioneering first-step platform towards a closed-loop diaphragm pacer regulation model
- Programmed embedded C firmware on RP2040, integrating multicore processing and communication protocols (I2C, UART, SPI) to handle IMU and CO2 sensor inputs with SD-card logging to synchronize capture at 100 Hz
- Guided team decision to implement backup through-hole Printed Circuit Board (PCB), reducing power usage by 20+ hrs and avoiding 15+ weeks of delays caused by failed high-fidelity iterations
- Partnered with clinicians to define clinically relevant features and applied signal processing algorithms on IMU and CO2 inputs, validating device feasibility while testing early feedback models

Continuous Live Web Streaming Device

Evanston, IL

Course Project for ELEC_ENG 326: Electronic System Design

Jan 2025 – Mar 2025

- Collaborated in a 2-person team to present a full-stack prototype within 10 weeks, with custom Eagle PCB, ESP32 web streaming, and 3D-printed enclosure
- Developed SAM4S MCU firmware in C integrating camera and ESP32 via SPI/UART, enabling 320x240 live stream

SKILLS

- **Programming & Tools:** Python, Embedded C, MATLAB, Git, CUDA, Adobe Illustrator/Photoshop/Premiere Pro, ImageJ
- **Hardware & Systems:** Raspberry Pi/Arduino/ESP32, Eagle PCB Design, CAD/Prototyping, Soldering, Signal Processing
- **ML Frameworks & Libraries:** MMPose, TensorFlow, PyTorch, OpenCV, scikit-learn, NumPy, Pandas
- **Languages & Interests:** Turkish (Native), English (Native), German (A2), Volleyball (HS Captain), Golf, Skiing, Tennis